MDM1 Group Project 13, Group 13

Battery Lifespan: An Applicable Model of Mobile Charge Management

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September 27, 2025

Abstract

LG Electronics have found that nine out of ten people experience 'low battery anxiety'[1]. This report aims to propose a mathematical model to describe the state of charge(SOC) of the mobile phone battery over time to reduce people's low battery anxiety.The model integrates Coulomb counting with voltage-based state of charge (SOC) estimation, using a 5th order polynomial to approximate Lithium-ion discharge curve. There are two models presented. The first model employs differential equations to track the SOC by decomposing the net current into charging currents, while the second model focuses on integrating voltage equations and Coulomb counting, suing the RC discharge equation to calculate current and SOC. Additionally, an application is proposed that using these models to monitor usage and predict remaining battery life. The combination of the mathematical models, practical application and discussion of battery management aims to improve people's experience and reduce anxiety around mobile phone's battery lifespan.

1 Introduction

In the digital age, mobile phones have become essential tools for communication, entertainment, and productivity. Research indicates that more than 13 million Britons experience significant stress when they are out of mobile phone contact. The primary reason for this attachment is the need to stay connected with friends and family, emphasizing the fundamental social connections facilitated by mobile phone usage[2].

This dependency has given rise to a modern phenomenon known as "Nomophobia". It is defined as the fear or anxiety of not having a working mobile phone[3]. The reason for not having a working mobile phone is usually due to a low battery.

This report aims to propose a mathematical model to describe the state of charge of the mobile phone battery over time in the form of using a differential equation. We will explore the battery lifespan taking into account the different states of the phone (charging, standby, in use). We will also perform a numerical integration and analyse the properties of its solution as the problem's relevant parameters vary.

2 Initial Assumptions

Phones are very complex devices - their charge and discharge rate are what we are tracking, but this can be changed drastically depending on what you're using your phone to do, and what situation you personally are in: for example, using your phone in the London underground on a hot summer day, while you run a graphics heavy game is a very different scenario to browsing Instagram.

This section will explain the process of making our model, and what assumptions we settled on keeping vs leaving out. A good initial model is not too complex, however, some variables are still very important to keep track of.

2.1 Temperature

It is apparent in our sources that the Temperature of the phone itself is linked to the battery drain of a phone[4].

The higher temperatures mean a phone has to work harder to stay cool and by extension, work as efficiently, since most devices have build in systems that stop the device if it gets too hot[5].

However, temperature isn't directly causing a phone to lose charge quicker: the loss of charge is a side effect of fans having to run faster, and more often than not, higher cpu/gpu intensity apps being used. This is why we have decided to model different scenarios.

2.2 Scenarios

A phone acts differently depending on what it is doing.

A quick assumption could be that a phone that is entirely powered off and a phone that is turned on will lose battery at different rates.

The first being the natural charge degradation rate of an uncharged battery (about 2 - 3 percent per month left unattended[6] and the second being based on the energy consumption of the phone. This is the most simple equations that can be made:

$$\frac{dQ}{dt} = Q_{\max} - (1000Wt/V)$$

Where Q is the charge of the phone in mAh, $Q_{\max}$is the maximum charge in mAh (milliAmp hours), W is the average Wattage used by the phone, t is how long the phone is being used for and V is the voltage of the phone battery.

Using only these two models however is not enough. The first flaw is that no acknowledgments have been made for power going into the phone. Many people will use their phone while it is on charge. Along with this, it is not reasonable to assume that a person will fully power off their phone when it's not in use, and it's also not reasonable to assume that the Voltage is constant. Voltage in a phone is variable, usually going from about 4.5V for a phone that is fully charged and around 3.2V for a phone that is on low battery[7].

This is also where temperature can be seen effecting the curve directly - there is a range of optimum temperatures (around 20ºC to 65ºC, as seen in[6]), through which the Lithium Ion battery will have the lowest internal resistance - a desirable trait in a phone because it means there are less drastic fluctuations in voltage[8].

The normal working range of temperatures for a phone is around 0ºC to 35 ºC, which is mostly inside of the range mentioned above. This means a simple model can safely ignore the effect of internal resistance on the voltage of the phone, meaning a typical Li-ion discharge rate can be used (see 2.

2.3 The Charging Equation

Lithium Ion batteries charge up to a certain point under constant current (CC) called constant current phase, after an SoC of 70-80 it reaches the constant voltage phase (CV), where voltage is held at a maximum limit and the current exponentially decreases to prevent overcharging which damages the battery. This phase takes longer due to exponential decline of the current. This means that for a portion of the charging time, they will charge at varied voltages (a faster rate of charge) before switching over to a constant voltage rate to finish the charging.

2.4 The Discharging Equation

Lithium-Ion batteries typically discharge in 3 stages providing there is a constant load/constant current; a beginning stage, a mid stage and an end stage. During the beginning stage, voltage will

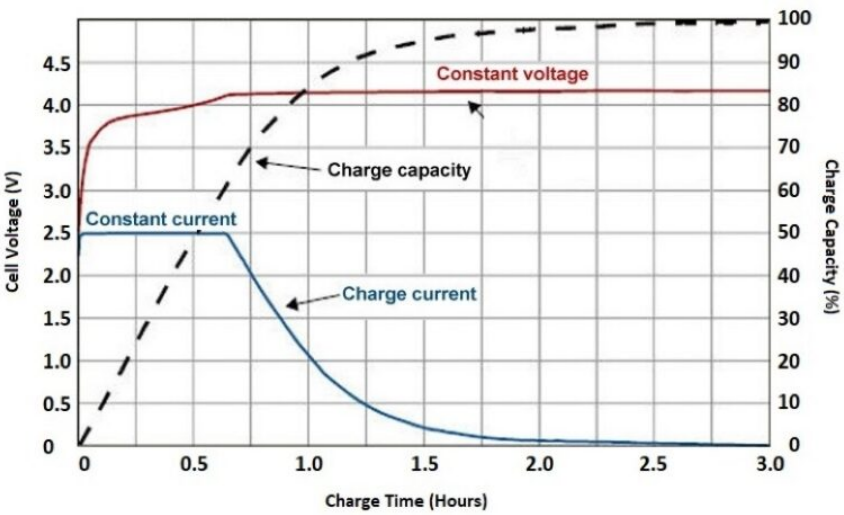


Figure 1: This is the charge curve for a typical Lithium Ion Battery - the kind found in our phones. It uses the constant charge, constant voltage (CC CV) method in order to protect the battery[8].

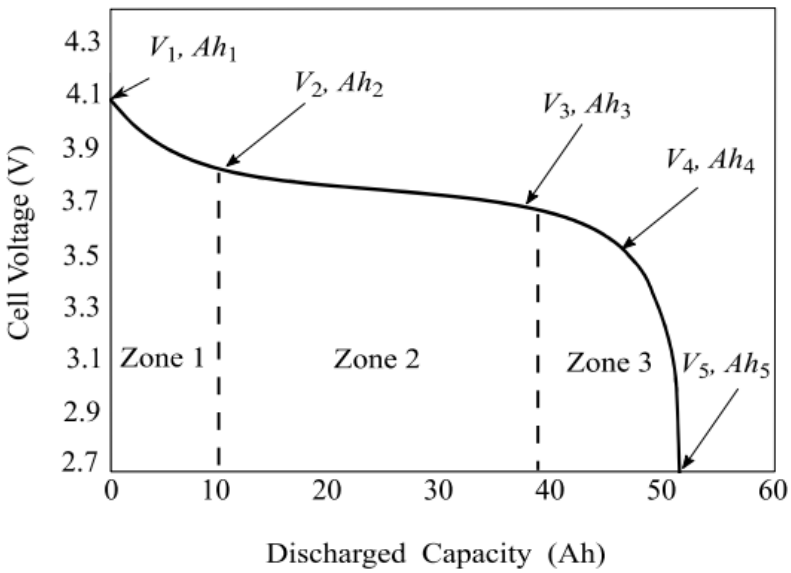


Figure 2: This is the discharge curve for a typical Lithium Ion Battery - the kind found in our phones. [9]

typically start at 4.2V and drop off rapidly to around 3.8V depending on the battery cell. This is when the mid stage starts. The mid stage is when the least voltage drop off occurs called an ohmic quasi-linear drop-off, voltage flattens out and remains consistent, providing a steady stream of power to the device. At around 3.5V the battery enters the end stage where the voltage will fall off rapidly again, this is similar to the rate it falls off at during the beginning stage. After battery voltage reaches the cut-off voltage at around 3.2V the phone will reach 0 and power off. This is put in place to prevent battery damage, as electrochemical processes internally start to become unstable below 3.2V as well as lithium build-up around the anode starts to occur. When modelling SOC we have to consider specific voltages to represent 0 and 100 so that battery damage can be prevented.

3 Methods and Results

3.1 How we could model Voltage Discharge and Charge

There are many models that can be used to model a discharge curve for a lithium-ion battery due to the complexity of the curve, we have decided on modelling the curve using a 5th order polynomial. This is the minimum order of polynomial required to model such a curve[9] . To improve accuracy, voltage drop-off due to internal resistance is also included in the model using the load current. This model can be used for any Lithium-ion discharge curve, however coefficients need to be determined from a MATLAB curve approximation.

$$V(t) = a_5t^5 + a_4t^4 + a_3t^3 + a_2t^2 + a_1t + a_0 - R_{INT}(i - i_T))$$

We will model the voltage using a 5th order polynomial. The voltage discharge curve for a Lithium Ion battery is almost always the same shape (not accounting for large discharges or faults in the battery), meaning a simple model can be made that uses the maximum and minimum voltage against the time. GSMArena gives a benchmark for the Samsung S25 Ultra of 10 hours and six minutes for their battery test on heavy usage (gaming)[10].

This means that the domain of our voltage discharge curve will be between zero and 10hrs6 minutes.

The rated voltage of the Samsung S25 ultra battery is a range between 4.2V and 3.7v - this gives us the positive range of our graph. Anything negative should be ignored.

The curve for voltage when using CC-CV charging is different to the discharge curve, it doesn't have the signature quasi-linear drop-off. This means we can model volage using a simple exponential equation during the CC phase, similar to those found in capacitors.

$$V(t) = V_{min} + (V_{max} - V_{min})(1 - e^{-\frac{t}{\tau}})$$

Where Vmax - Vmin represents the voltage range which will be between 1.0 and 1.2. This only works for CC as at CV, voltage remains at Vmax. τ Represents the time constant or RC which is the internal resistance and the capacitance of the battery itself. When in a phone load resistance and capacitance increases so RC increases. Both these models are only for Voltage and while for discharge you can base SOC directly off Voltage, for CC-CV charging, SOC still increases while voltage is kept constant, due to an increasing current, so basing SOC off that is harder.

3.1.1 How is battery percentage measured?

Modern phones use a combination of four methods to calculate their battery percentage - voltage measurement, coulomb/charge counting, impedance tracking and temperature compensation[11] . Snyder states that the two most common formulas for giving us a state of charge (SOC) are as follows:

The first uses Coulomb Counting:

$$SOC_{new} = SOC_{previous} - \frac{I \cdot \Delta t}{C_{rated}}$$

Where SOC is the state of charge, I is the current and C rated is the maximum mAh as stated on the battery.

The second uses Voltage measurement:

$$SOC = \frac{V_{measured} - V_{min}}{V_{max} - V_{min}} \cdot 100\%$$

3.2 Model 1

In this model, the actual current under various conditions is calculated by decomposing the total current, enabling a piecewise evaluation of the charging rates corresponding to different scenarios.

3.2.1 State of Charge (SoC) Definition

In this model, we let:

$$\mathrm{SoC}(t) = \frac{Q_{\max}}{Q(t)}$$

Where:

Q(t) is the instantaneous battery charge, Q(max) is the maximum available charge. So SoC(t) varies between 0 and 1.

3.2.2 Differential Equation

We can start with

$$Q = I \cdot \Delta t$$

By differentiating this equation, we obtain

$$\frac{dQ}{dt} = I_{\mathrm{net}}(t),$$

Where:

$I_{\mathrm{net}}(t)$ is the net current into the battery (positive) or out of the battery (negative). In terms of SoC,because $Q(t) = Q_{\max}\,\mathrm{SoC}(t)$, we have:

$$\frac{d\,\mathrm{SoC}(t)}{dt} = \frac{1}{Q_{\max}}\frac{dQ}{dt} = \frac{1}{Q_{\max}} I_{\mathrm{net}}(t),$$

3.2.3 Decompose current $I_{net}(t)$

Since $I_{\mathrm{net}}(t)$ consists of both input and output current components, we directly decompose $I_{\mathrm{net}}(t)$ into input and output currents to facilitate subsequent analysis of the charging and discharging rates of the battery under different operating conditions.

$$I_{\mathrm{net}}(t) = I_{\mathrm{in}}(t) - I_{\mathrm{use}}(t)$$

$I_{in}(t)$:the charger current; $I_{use}(t)$: the current used

3.2.4 Load and Charging Current

Load Current:in this context, the current I is determined using the power equation P = IV.

$$I_{\mathrm{use}}(t) = \frac{P(t)}{V\big(\mathrm{SoC}(t)\big)}$$

where the instantaneous power consumption of the phone is P (t) and the battery voltage is V(SoC(t)). P (t) could be piecewise constant for "in use" vs. "standby", however, to simplify subsequent calculations, we refer to the existing literature and adopt a fixed power value of P=5.9W[12].

Charging Current:

CC: For SoC below some threshold (battery voltage less than 4.2V), we might approximate

$$I_{\mathrm{in}}(t) = I_{\mathrm{CC}}$$

where I_{CC} is a constant.

CV: when V(SoC(t)) = 4.2V, the charger keeps voltage constant at V_{max} The current typically decays as

$$I_{\text{in}}(t) = \frac{V_{\max} - V\big(\text{SoC(t)}\big)}{R_{\text{int}}}$$

where R_{int} denotes the equivalent internal resistance

3.2.5 Differential Equations in Different Conditions

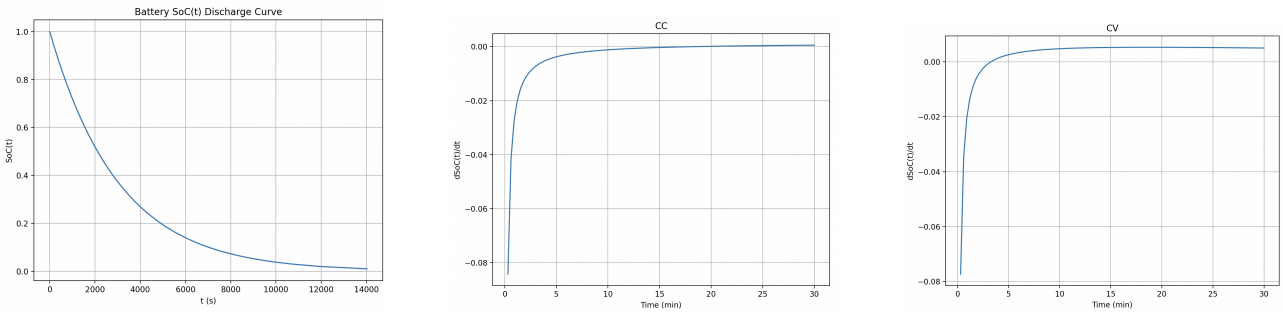
By combining the above equations, we obtain:

$$\frac{d\,\text{SoC}(t)}{dt} = \frac{1}{Q_{\max}}\Big[I_{\text{in}}(t) - \frac{P(t)}{V\big(\text{SoC}(t)\big)}\Big].$$

By substituting the expressions for $I_{in(t)}$ under different conditions, we can derive the final form of the equations corresponding to each respective state.

$$\frac{d\,\text{SoC}(t)}{dt} = \begin{cases} -\dfrac{P(t)}{Q_{\max}\,V\big(\text{SoC}(t)\big)}, & Discharging \\ \dfrac{I_{\text{CC}}}{Q_{\max}} - \dfrac{P(t)}{Q_{\max}\,V\big(\text{SoC}(t)\big)}, & CC \\ \dfrac{V_{\max} - V\big(\text{SoC}(t)\big)}{Q_{\max}R_{\text{int}}} - \dfrac{P(t)}{Q_{\max}\,V\big(\text{SoC}(t)\big)}, & CV \end{cases}$$

where $Q_{max}$is 5000mAh, charging: 45W wired,65% in 30 min[13]



(a) Discharging (b) CC (c) CV

Figure 3: SoC(t)graphs under different conditions

3.3 Model 2

Having the voltage equation allows us to finally combine all of our information into one final model, starting with the SOC equation using Coulomb counting.

$$SOC_{new} = SOC_{previous} - \frac{I \cdot \Delta t}{C_{rated}}$$

$$\rightarrow \frac{SoC_{new} - SoC_{previous}}{\Delta t} = -\frac{I}{C_{rated}}$$

To use this equation, we need a value for I. This is our method for finding I.

3.3.1 Constants and values

We can find I through the use of three main equations:

$$P = I_2 \cdot V_2$$

Where:

- Power is a constant (found by taking the average wattage of a phone for a certain task - for our example, we will be using a benchmark test for a CPU heavy game which gives average wattage as 5.9W[12]).
- I_2 is the Amperage of the phone at time t. this is what we are trying to find.
- V_2 is the Voltage at time T which will be found through the use of the equation mentioned in 3.1

$$V_2 = I_2 \cdot R_2$$

Where R_2 is Resistance of the battery. This is created by the load of the CPU and is a value we do not have without sensors on the battery itself.

And finally, the RC discharge equation[14]

$$V_2 = V_0 \cdot e^{\frac{-t}{R_2 C}}$$

Where

- V_0 is the initial voltage of the battery - in our example case, 4.2V
- t is the time at the point we are testing. This is useful if you wanted to predict SOC after 30 minutes of use.
- C is the capacitance of the battery, which is a constant.

3.3.2 How the equation is formed

Starting with the basic electricity equations and the RC discharge equation[14] , the process is numbered as follows:

$$I_2 = \frac{P \cdot e^{tPV_2{}^2 C}}{V_0}$$

This equation gives us the Amps at any given time. We can differentiate it to create our model's differential equation:

$$I = Ae^{BtV_2^2}$$

$$\rightarrow \frac{dI}{dt} = ABV_2^2 \cdot e^{BtV_2^2}$$

Where $A = P/V_0$, and $B = P \cdot C$

This gives us a value for I at any point on a charging or discharging curve for voltage against time. To do this, you can substitute any curve in terms of V against t into the spaces for V_2. These are curves that can be approximated in many different ways. For example, we can see in the charge curve 1 that the "CC" portion of the voltage curve is an exponential shape until the battery is about 70% full. Our benchmarks for our phone show that this takes about 30 minutes of charging. So, a monotonic

equation that approximates this shape is: $V_2 = 4.2 - 1.7e^{-0.0946t}$. This can be substituted directly in place of V_2, giving us:

$$\frac{dI}{dt} = AB(4.2 - 1.7e^{-0.0946t})^2 \cdot e^{Bt(4.2-1.7e^{-0.0946t})^2}$$

3.3.3 SoC Differential Equation

Meanwhile, by combining the differential equation of current with the state of charge (SoC) equation derived using Coulomb counting, we can obtain the differential equation governing the SoC.

$$\frac{d\,\mathrm{SoC}(t)}{dt} = -\frac{1}{C_{\text{rated}}}\left[I(0) + \frac{P}{V_0}\left(e^{2\,P\,C\,V_2^2\,t} - 1\right)\right].$$

$$\rightarrow \frac{d\,\mathrm{SoC}(t)}{dt} = -\frac{1}{C_{\text{rated}}}\left[I(0) + A\left(e^{2BV_2^2\,t} - 1\right)\right].$$

In this formulation, I(0), A, B, and C_{rated} are treated as constants, which differ depending on the specifications of each smartphone model. As such, we refrain from conducting explicit numerical evaluations here. Nevertheless, the proposed equation remains applicable for estimating the charging rate under both Constant Current (CC) and Constant Voltage (CV) charging regimes.

4 App proposal

According to the LG "Low Battery Anxiety" survey, 9 out of 10 people experience a sense of distress or discomfort when their phone battery reaches low levels, typically around 20 percent or below. This is known as low battery anxiety; this leads people to alter their daily routine whether that be carrying multiple chargers, limiting their phone usage, or not participating in events as their phone is too low on battery, such as going to the gym. To address this issue, we propose the development of an app to mitigate low battery anxiety by making predictions based on data and previously defined differential equations. Our battery will include real-time battery monitoring, predicting estimated life span of battery remaining, and recommending power saving actions.

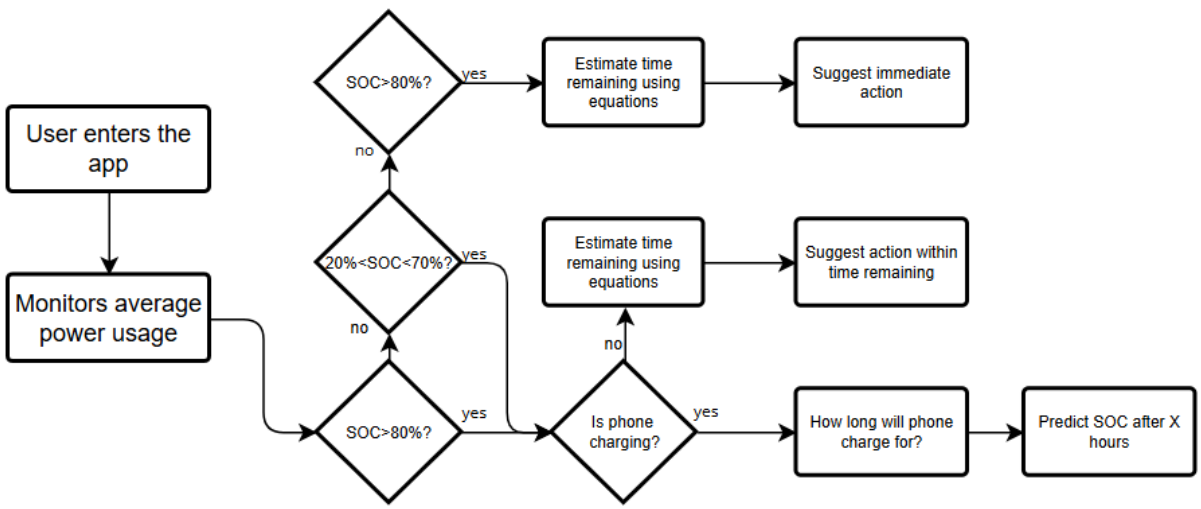


Figure 4: This is a for a flowchart representing the functionality of the proposed app

The app begins by monitoring the user's average power usage and tracking the current State of Charge (SOC). The objective of the app is to alleviate low battery anxiety by providing an estimation of the

battery life. Based on this estimate the app will recommend an appropriate action based on this battery life to extend it, such as turning on power saving mode or turning off Bluetooth and location services.

5 Discussion and Conclusion

5.1 Merits and Limitations of models

5.1.1

The latter model proposed in this report has the following advantages:

Model 1 predicts the state of charge (SoC) of a battery using differential equations that take into account the net difference between charge and use currents. It models the voltage by a 5th order polynomial that reflects the discharge curve of the Li-ion battery and aims to provide SoC prediction for different usage scenarios.The advantages of the model are that it is comprehensive, takes into account both voltage and current variations, is suitable for a wide range of mobile phones, has low computational requirements and is easy to implement and tune the parameters. Model 2 integrates the voltage equation and the RC discharge model to predict current and SoC, providing a more detailed simulation of battery behaviour, suitable for complex usage scenarios. Its advantages are that it is more accurate, taking into account the dynamic changes of the battery, suitable for predicting high-intensity tasks, more flexible, and the voltage profile can be adapted to different situations.

5.1.2

However, this model also has some limitations: Model 1 is based on differential equation tracking of SoCs and is suitable for various mobile phones, but accuracy is dependent on current measurements. Its drawbacks are that it may not be applicable to some brands, accuracy is limited by the accuracy of current measurements, and voltage polynomials may not be accurate enough in some battery states. Model 2 is more detailed, integrates voltage and current dynamics and can be more accurate, but is complex and may not be fully adapted to Li-ion battery behaviour, requires more parameters to the point of high computational resource requirements, relies on specific data, may not be generic and has limited applicability to certain brands.

6 Battery Management

Battery management is crucial in preventing low battery anxiety by acting in maintaining the efficiency, longevity and safety of battery and battery-powered device. Effective battery management involves monitoring and optimizing various parameters such as state of charge, state of health and temperature. Here are some key practices:

1. Avoiding Overcharging: Ensuring that the device is unplugged after reaching full charge. This prevents excessive voltage stress, which can damage the device and shorten battery life.

2. Maintaining battery health: When using a new phone for the first time, full drain the battery until it power off before charging it. This is because new devices often come with a 'pre-charged' or 'shallow' battery level from the factory, which may not reflect its actual capacity. Charging it after a full discharge helps to calibrate the battery meter for better accuracy.

3. Partial charging: Charging the battery to around 80%-90% instead of 100% can reduce wear and extend the overall lifespan.

4. Temperature management: Keeping the device at moderate temperatures during charging can prevent degradation, as high and low temperatures can accelerate chemical breakdown.

7 Conclusion

We introduce nomophobia, the fear of being without mobile phone contact, often due to battery depletion, and aim to develop a mathematical model to predict battery SoC over time, considering different phone states and usage scenarios. We use two models, Model 1: Uses a differential equation to track SoC, decomposing the net current into charging and usage components. Model 1: Uses a differential equation to track the SoC, decomposing the net current into charging and usage components. Voltage is modelled with a 5th order polynomial to reflect the lithium-ion discharge curve, providing a detailed view of battery behaviour under the influence of the battery. providing a detailed view of battery behaviour under different conditions.Model 2: Builds on Model 1 by integrating voltage equations and Coulomb Model 2: Builds on Model 1 by integrating voltage equations and Coulomb counting, using the RC discharge equation to predict current and SoC, with examples such as the battery performance of the Samsung S25 Ultra. We propose an app to alleviate low battery anxiety by monitoring real-time usage, predicting remaining life, and recommending actions such as turning on power We propose an app to alleviate low battery anxiety by monitoring real-time usage, predicting remaining life, and recommending actions such as turning on power saving mode based on the models' predictions. show the model predicts SoC changes under different scenarios via numerical integration, analysing voltage and current dynamics. Benefits include comprehensive modelling and adaptability, but limitations point to potential inapplicability to brands such as iPhones or Nokias due to variations in battery data.Using mathematical modelling, battery science and practical applications, we present two models and application scenarios that aim to reduce user anxiety through accurate predictions and practical advice. Despite the limitations of the models, the combination of apps and battery management tips can significantly improve the user experience.

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